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Multi-step verification of the Copenhagen atomics molten salt experiment radioactive inventory at the Paul Scherrer Institute using OpenMC, Serpent, and EQL0D

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The molten salt experiment (MSE) is a 30-day 1 MW (thermal) molten salt criticality experiment designed by Copenhagen Atomics that will be conducted at the Paul Scherrer Institute (PSI) Swiss research center. A detailed analysis of the radioactive inventory available during the operation and after shutdown has been carried out for future estimates of potential source term release in case of undesired accidents using the Monte Carlo transport and depletion code OpenMC. To benchmark the radioactive inventory obtained through the OpenMC burn-up calculation (version 0.15.2), code-to-code comparison with the Serpent Monte Carlo and depletion code (v2.2.1), as well as with Serpent coupled to the EQL0D depletion module developed at PSI, were performed. All of the utilized codes support neutron transport on CAD-based geometries and allow continuous material reprocessing during depletion. This feature is essential for modeling the MSE concept, where gaseous fission products are continuously extracted from the circulating fuel salt. For the present verification study, three models of increasing geometric and physical detail were developed: (1) a nested hollow sphere model in constructive solid geometry (CSG), with the scope of aligning basic settings beyond geometry; (2) a geometrically simplified CAD model of the Copenhagen Atomics Onion Core, mainly utilized for testing the integration of CAD models within the above depletion codes; (3) the same simplified CAD model but with the implementation of reactor fluid flows from the inner active core to the outer inactive region, mainly to account for the effect of out-of-core decay. The results were compared to benchmark the original OpenMC model. Particular attention was given to the end-of-cycle (EOC) isotopic inventory, with detailed comparisons for key radionuclides and total radiotoxicity.

KEYWORDS

Monte Carlo neutron transport, burnup calculations, molten salt reactor, OpenMC, Serpent, EQL0D, CAD, CSG

1 Introduction

The Copenhagen Atomics (CA) Molten Salt Experiment (MSE) is a molten salt test reactor currently under development by CA in collaboration with the Paul Scherrer Institute (PSI), which will host the experiment on its premises. The MSE is designed as a low-burnup test, planned to operate for 1 month at 1 MW of thermal power. This limited operational scope enables the project to pursue a low-hazard license, helping to accelerate and reduce uncertainties in the regulatory approval process.

Despite its low power and short duration, a comprehensive safety assessment is still required. This includes a detailed accident analysis to estimate the potential source term in the event of severe accidents that could lead to radiological releases to the public.

To accurately estimate the source term, a combination of computational tools is required: primarily a depletion code to calculate the radionuclide inventory and a dispersion code to model the environmental release and transport of radioactive materials.

In this context, OpenMC (Romano and Forget, 2013) is a Monte Carlo neutron transport code developed at the Massachusetts Institute of Technology (MIT) that includes an integrated burnup solver for coupled depletion simulations. Starting with version 0.15.2, OpenMC's depletion solver was extended to support continuous material reprocessing, enabling the modeling of online operations such as fission product removal and fuel salt refueling. These features are essential for simulating the MSE, where volatile fission products are continuously extracted from the fuel salt and directed towards an off-gas system (OGS). The extension was implemented by incorporating additional terms into the Bateman equations to account for material inflow and outflow during depletion.

Similar capabilities have also been introduced in other codes. Serpent implemented continuous reprocessing support, starting from version 2.1 (Aufiero et al., 2013). EQL0D, developed at PSI (Hombourger et al., 2020), is a Matlab-based wrapper for Serpent, written with the primary goal of predicting the evolution of the fuel composition in molten salt reactors during burn-up. It includes advanced routines for online material transfers not only based on user-defined transfer rates but also on conservation quantities, reactivity control, and salt oxidation-reduction potential control. The SCALE code system has introduced a new MSR-specific depletion module, TRITON-MSR (Lo et al., 2022), while *ad hoc* depletion solvers like ADDER (Anderson et al., 2022) and SaltProc (Rykhlevskii et al., 2019) have been developed specifically for MSR applications. These tools can be coupled with Monte Carlo transport codes to perform comprehensive simulations of MSR fuel cycles, including detailed fuel reprocessing schemes.

The present article presents a detailed description of the methodologies adopted by three different depletion codes—OpenMC, Serpent, and EQL0D—to model molten salt reactors (MSRs). A validation case study is introduced, consisting of three sequential models of increasing complexity, each exploring different aspects of core geometry and reprocessing behavior.

The first model uses a constructive solid geometry (CSG) simplified representation of the CA Onion Core, with a single reprocessing scheme applied to volatile fission products. It is proposed in order to not source any geometry-related discrepancies and at the same time to be representative of the actual reactor design. This model is proposed to fine-tune the basic settings of the transport and depletion codes being considered.

The second model introduces a more realistic CAD-based geometry of the Onion Core, allowing for a deeper investigation of potential discrepancies between Monte Carlo transport solvers, particularly due to the differences in how OpenMC and Serpent handle CAD geometry conversion, despite using the same depletion settings.

The third and most advanced model builds upon the CAD geometry and incorporates more detailed reprocessing schemes,

providing a comprehensive test of the depletion solvers' capabilities in simulating realistic MSR fuel cycle behavior.

2 Methodology

Molten salt fluid-fueled reactors are associated with a variety of reprocessing schemes designed to efficiently manage both fuel and waste materials. These schemes are defined by the techniques used to remove fission products and recover fissile material, and they play a critical role in shaping the fuel cycle of a given reactor design. In typical MSR operation, elements that do not form stable compounds are generally easier to remove. Their behavior and removal mechanisms have been extensively studied, particularly building on the operational experience of the Molten Salt Reactor Experiment (MSRE) conducted at Oak Ridge National Laboratory (ORNL) in the 1960s. For instance, noble gases tend to bubble out of the molten salt and diffuse toward void regions or into off-gas systems, while noble metals often plate out onto metallic surfaces within the fuel circuit.

To mathematically represent the removal of individual nuclides, the process can be modeled using an exponential decay law, analogous to radioactive decay. In this analogy, the removal is governed by an artificial decay constant, here defined as the transfer rate coefficient t , which has the same units as a physical decay constant. From a physical perspective, this coefficient is often more intuitively expressed as the reciprocal of a cycle time—the time required to circulate and process the entire salt inventory at a specified flow rate and separation efficiency.

To accurately simulate the continuous removal or feeding of materials during reactor operation, this transfer rate term must be incorporated into the Bateman equation. The standard Bateman equation accounts only for nuclear reactions and radioactive decay, and by adding the transfer rate term, can be written as:

$$\frac{dN_i(t)}{dt} = \underbrace{\sum_j \left(\int_0^\infty N_j(t) \phi(E, t) \sigma_{j \rightarrow i}(E, t) dE \right)}_{\text{Reaction rate term}} - \underbrace{\sum_j \left(\int_0^\infty N_i(t) \phi(E, t) \sigma_{i \rightarrow j}(E, t) dE \right)}_{\text{Transfer rate term}} + \underbrace{\sum_j N_j(t) \lambda_j b_{j \rightarrow i}}_{\text{Decay term}} - N_i(t) \lambda_i - N_i t_i + \sum_j N_j t_{j \rightarrow i} \quad (1)$$

where $N_i(t)$ is the number of atoms of nuclide i , $\phi(E, t)$ is the neutron flux as a function of energy and time, $\sigma_{i \rightarrow j}(E, t)$ is the microscopic capture cross-section from nuclide i to j , λ_i is the decay constant of nuclide i , and $b_{j \rightarrow i}$ is the branching ratio for nuclide j decaying to nuclide i . The proportional transfer rate term models the continuous transfer of nuclides as a function of the transfer rate coefficient t_i for nuclide i and $t_{j \rightarrow i}$ for nuclide j transferring to i , which describes the rate and efficiency of the reprocessing scheme adopted. The negative sign before the transfer rate term is just a convention adopted here, where a positive and negative t represent a removal and feed, respectively. Reaction rates obtained from the transport solver, together with decay chain data, are passed to the depletion solver to construct the burnup matrix $\mathbf{A}$, which is subsequently used for the integration of Equation 1. In this study, we adopt the OpenMC terminology, where the additional removal or feeding term is referred to as a

“transfer rate”. In contrast, Serpent and EQL0D generally do not explicitly use this terminology but rather treat transfer processes as a form of generalized artificial decay.

2.1 Transfer rates in OpenMC

The transfer rate coefficient t acts as an artificial decay constant, representing the rate at which a nuclide is removed from or added to the system due to reprocessing operations. It follows from Equation 1 that, when nuclides are removed without being tracked, the transfer rate coefficient t_i is subtracted from the corresponding diagonal term of the Bateman matrix. Consequently, the i^{th} diagonal entry takes the form $-\phi_i \sigma_i - \lambda_i - t_i$.

Alternatively, if nuclides belonging to one material get transferred to or from another, the two materials become coupled as well as their burnup matrices. The term $\sum_j N_j t_{j \rightarrow i}$ from the transfer rate term of Equation 1 is used to construct a diagonal transfer matrix $\mathbf{T}_{ij}$ that provides the coupling link between the i^{th} and j^{th} materials, where the two indices represent the elements receiving and losing materials, respectively.

The burnup and transfer matrices are then assembled to form a block matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{pmatrix} \ddots & \cdots & \cdots & \cdots \\ \vdots & \mathbf{A}_{ii} & \mathbf{T}_{ij} & \cdots \\ \vdots & \mathbf{T}_{ji} & \mathbf{A}_{jj} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad (2)$$

with $A_{ii} = B_{ii} - T_{ji}$ and $A_{jj} = B_{jj} - T_{ij}$, where B_{ii} and B_{jj} represent the i^{th} and j^{th} burnup matrices that include all depletion terms, while T_{ij} and T_{ji} represent the transfer matrices including the transfer rate terms only. For instance, we consider the simple case of two depletable materials and one transfer defined from material 1 to material 2, so that $\mathbf{T}_{21}$ needs to be defined. The system for solving will then result as follows:

$$\frac{d}{dt} \begin{pmatrix} \vec{N}_1 \\ \vec{N}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{T}_{21} & \mathbf{A}_{22} \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} \vec{N}_1 \\ \vec{N}_2 \end{pmatrix} \quad (3)$$

With this implementation, the cyclic transfer of materials between the in-core, neutron irradiated regions, and the out-of-core, decay only regions, within a single calculation step, as a function of the fluid flow rate, can emulate the material circulation, characteristic of MSR fluid loops.

2.2 Transfer rates in Serpent

An extended version of Serpent capable of removing and feeding materials during depletion was firstly implemented for the extensive modeling of MSRs (Aufiero et al., 2013). While features such as criticality control are unavailable in the version used in this benchmark (v2.1.1), among the selectable Serpent material reprocessing modes, the transfer through an artificial decay constant is selectable and obeys Equation 1 (Leppänen et al., 2025).

While extensive documentation on the implementation of material transfers is not available, several tests were performed to understand that transfer loops as described in Equation 2 are not supported; that is, the code cannot propagate material changes in cyclic transfers as OpenMC. The code is designed to process transfers of materials only in linear chains (as the direct transfer in Equation 3) and therefore limits the use of artificial decay to simultaneously transfer materials from in-core active regions to out-core inactive regions and *vice versa*.

2.3 Transfer rates in EQL0D

EQL0D (Hombourger et al., 2020) is a Matlab-based wrapper for Serpent2 developed at PSI with the primary goal predicting the evolution of the materials composition in MSRs during burn-up. Several different refueling and reprocessing stream options are available, including continuous or batch-wise addition or the removal of selected isotopes, as well as the transfer of isotopes from one material to another (as fission gases from the fuel into an off-gas system). Conservation modes, criticality control, and molten salt reduction-oxidation control are also available. Among the EQL0D material reprocessing modes, the transfer through an artificial decay constant is selectable.

The implementation of material transfer through artificial decay is implemented similarly as in OpenMC, with the difference that in OpenMC the Bateman system is solved for the number of atoms N_i , while in EQL0D it is solved for the atomic densities d_i , requiring a volume re-normalization in the transfer matrices $\mathbf{T}$ in Equation 3, delivering

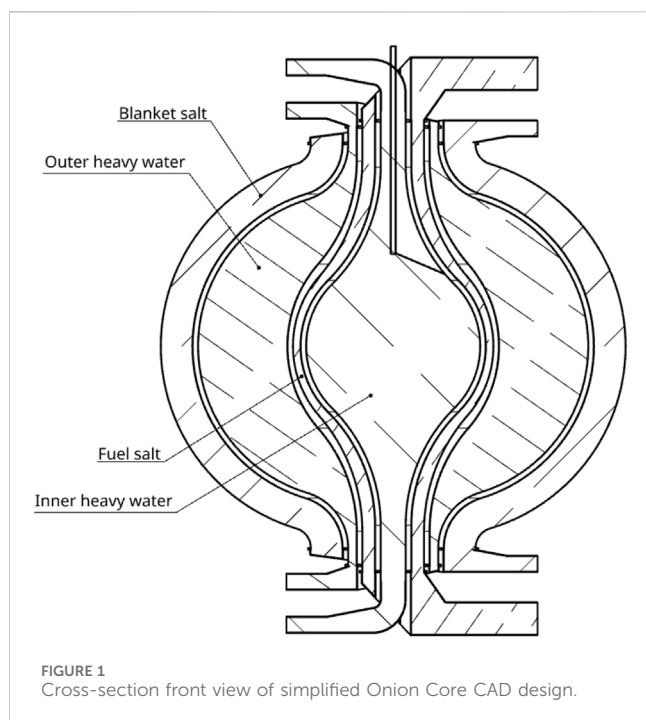
$$\mathbf{A} = \begin{pmatrix} \ddots & \cdots & \cdots & \cdots \\ \vdots & \mathbf{A}_{ii} & \frac{V_j}{V_i} \mathbf{T}_{ij} & \cdots \\ \vdots & \frac{V_i}{V_j} \mathbf{T}_{ji} & \mathbf{A}_{jj} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad (4)$$

while the $\mathbf{A}$ matrices remain untouched.

While in OpenMC the artificial decay constant integrates both the processing time of the material and the nuclide transfer efficiency $t = \lambda \cdot \epsilon$, in EQL0D the artificial decay constant and removal efficiency can be selected as distinguished parameters. This allows a series of isotopes to be removed with the same material processing time while accounting for their different removal efficiency. In the example of fission gases, removal efficiency can indeed vary based on gas solubility, diffusivity, and buoyancy.

3 Onion core model description

The core of the CA Onion consists of several concentric spherical channels that circulate fuel and blanket molten salts from bottom to top, along with a heavy water moderator separated into two distinct regions. Figure 1 shows a cross-section of a representative, though simplified, CAD model of the actual Onion core. This model serves as the reference geometry for



the reactor simulations presented in this study. The constructive solid geometry (CSG) model serves as the first simplified step in the benchmark process. It consists of a series of hollow concentric spheres, with the sole exception of the innermost region (the inner heavy water moderator), which is modeled as a solid sphere. This configuration mirrors the dimensions of the more detailed CAD geometry (Figure 2), but it does not constitute an

exact replica; therefore, differences in the transport solver results are expected. The CAD model of the CA Onion Core was designed using the CAE tool OnShape. To enable particle transport simulations on CAD-based geometries, the approach used with OpenMC involves converting the CAD model into a discretized STL (stereolithography) surface mesh, where geometries are represented by a list of tiny triangles that build up to 3D shapes, using the open-source meshing tool CAD-to-OpenMC (Knudsen and Chierici, 2025) in conjunction with the DAGMC toolkit (Wilson et al., 2010), which allows for direct particle tracking on tessellated surfaces. The output is an h5m file, that can be input directly into OpenMC through the native support for DAGMC geometries. The meshing algorithm used in CAD-to-OpenMC is `stl2`, which uses the underlying library OpenCASCADE and is able to handle coincident surfaces. The meshing parameters used are the following.

- Min. mesh size [mm]: 0.1
- Tolerance [mm]: 0.001
- Angular tolerance [rad]: 0.2

On the other hand, to import CAD geometries of solids in Serpent, the `solid 1` card can be used, where unstructured meshes defining the domain's geometry can be imported. Similarly, the CAD files need to be converted into STL meshes. In the STL-based methodology, the cell search routine loops over the STL solids with adaptive search cell and defines the location's material with ray-tracking-based methods (Leppänen, 2022).

Altogether, the refinement of the STL geometry and the settings for the adaptive search geometry are parameters that potentially impact the transport calculation. Whenever performing code-to-

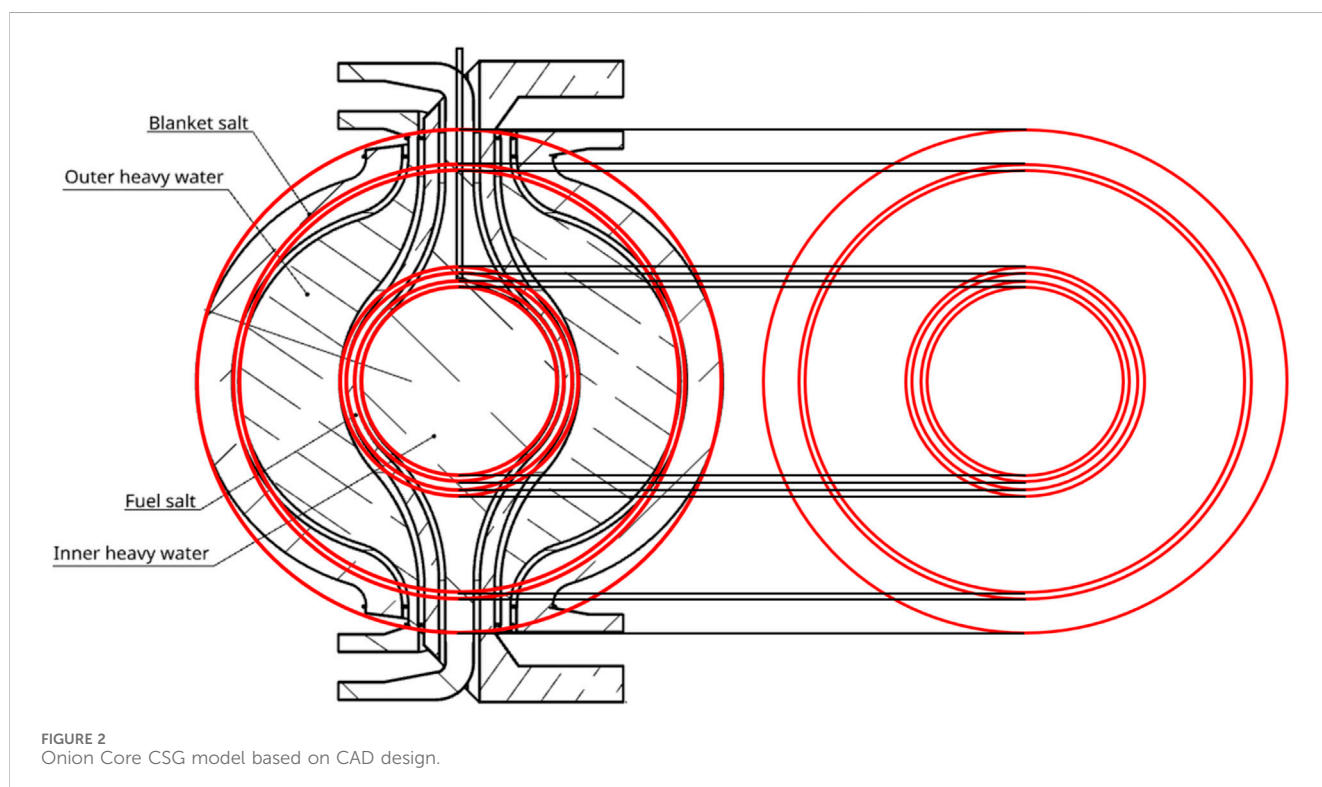


TABLE 1 Main reactor core parameters.

Parameter	Value	Parameter	Value
Thermal output	1 MW(th)	(D ₂ O) moderator temperature	26.8 °C
Core dimensions	Height ~ 2 m, width ~ 2 m	(D ₂ O) moderator density	1.104 g/cm ³
Fuel salt composition	LiF-UF ₄ (73–27 mol%)	(D ₂ O) moderator volume	3,000,000 cm ³
Blanket salt composition	LiF-ThF ₄ (70–30 mol%)	OGS temperature	26.8 °C
Fuel salt temperature	626.8 °C	OGS density	1.83 g/cm ³
Fuel salt density	4.905 g/cm ³	OGS volume	1,000,000 cm ³
Blanket salt temperature	626.8 °C		
Blanket salt density	4.626 g/cm ³		
Fuel salt volume	500,000 cm ³		
Blanket salt volume	3,000,000 cm ³		

code benchmarks, it is therefore relevant to compare the methods used to implement the CAD geometries and fine-tune the impacting input files and parameters. The STL meshing parameters are the following.

- Surf. Dev. [m]: 0.000563
- Norm. Dev. [deg]: 15.0
- Max. Edge Length [m]: 3.5546
- Aspect Ratio: 21.50

The adaptive cell search parameters are: MESH_SPLIT: splitting criterion for the adaptive search mesh (maximum number of geometry cells in search mesh cell), set to 10; MESH_DIM: number of levels in the adaptive search mesh, set to 4; SZ1: size of the search mesh at level 1, set to 4; SZ2: Size of the search mesh at level 2, set to 3; SZ3: size of the search mesh at level 3, set to 2.

The fuel salt used is an eutectic mixture of LiF-UF₄ (73–27 mol%), enriched to 4.95wt% ²³⁵U and 99.99wt% ⁷Li. The blanket salt consists of LiF-ThF₄ eutectic mixture (70–30 mol%), using the same ⁷Li enrichment as the fuel salt, although the MSE test will use natural lithium. The moderator is heavy water (D₂O) enriched to 99.9wt% ²H. The channel walls in contact with the heavy water moderator are made of zirconium alloy, while those enclosing the molten salts use stainless steel 316. A thermal insulation layer is present between the heavy water and molten salt channels, although is not included in the models. An off-gas system (OGS) is also defined, intended to collect gaseous fission products. It is modeled as having an initial volume of 1 m³ and is filled with argon gas at 300 K; however, it is not included geometrically in the transport model as it is unaffected by neutron flux. The main reactor core parameters used in the transport and depletion models are listed in Table 1. Note that the reported total volumes represent the available inventories in the system rather than their geometric volumes.

3.1 Transport and depletion settings

Properly aligning nuclear data for coupled neutronics-depletion comparisons across different codes can be challenging, requiring a

detailed understanding of the underlying computational procedures and the differences in implementation between codes. The guidelines provided in Romano et al. (2021) have been instrumental in verifying the necessary configuration settings. Below, the main parameters influencing the outcome of the comparison are briefly discussed.

3.1.1 Cross section data

The workflow for the construction of the neutron cross-section (XS) data used in the transport model consisted in utilizing all the available data in the ENDF/B-VIII.0 library and complementing them with missing nuclides sourced from JEFF-3.3. This combined library, developed at PSI, includes a total of 631 tabulated nuclides¹. All material temperatures are approximated to available incident neutron temperatures of 900K and 300K to not rely on different temperature treatment methods used by the different transport codes.

3.1.2 Transport solver

The neutron transport simulation was set up using 200 batches, 20 of which were inactive and 1.5×10^5 source neutrons per batch.

3.1.3 Depletion settings

The evaluation method for the burnup matrix exponentials is the incomplete partial fraction (IPF) CRAM for OpenMC and partial fraction decomposition (PFD) CRAM for Serpent, both of order 16 (Pusa, 2011; Pusa and Leppänen, 2010; Pusa, 2016). These two methods are similar and are expected to give very similar results. The time integrator scheme used for the comparison is a predictor integrator.

3.1.4 Power normalization

Reaction rate normalization is calculated, assuming constant total power and total heating in the system obtained from Serpent

¹ The list of additional nuclides from JEFF-3.3 is provided in the attached datasets.

constant fission heating values², to ensure that identical Q values are used between the different codes.

3.1.5 Branching ratios

Capture branching ratios are taken to be identical to the default constant spectrum-averaged branching ratios used in Serpent, calculated from energy-dependent branching ratios in the JEFF-3.1 activation file from a PWR flux spectrum³.

3.1.6 Decay and fission product yield data

Data for decay and fission product yields (FPYs) from ENDF/B-VIII.0 are used. Energy dependence of FPY is treated differently, depending on the code used. Serpent uses a linear interpolation for non-tabulated energies to calculate an effective FPY. OpenMC adopts a method similar to that of Serpent in which the average energy of fission events is tallied. An effective fission product yield (FPY) is then computed by linearly interpolating between tabulated FPY data using this average energy, commonly referred to as the “average fission yield method”. These two implementations should be similar but not fully equivalent and might give rise to small discrepancies.

3.1.7 Transmutation reactions

The OpenMC and Serpent transport solvers compute reaction rates by explicitly tallying quantities like $N_i \phi \sigma_{i \rightarrow j}$ of Equation 1 using a track-length estimator and subsequently collapsing them into a one-group reaction rate coefficient for each nuclide and reaction considered. A total of 30 transmutation reactions present by default in the Serpent depletion module have been used and tracked by all codes. Based on the initial inventory and the reaction channels considered, this corresponds to a reduced depletion chain of approximately 1800 nuclides.

3.1.8 Depletion settings

The MSE is depleted to 30MWd using a single depletion step. Wall materials are not set as depletable.

4 Models

4.1 Model 1

In the first benchmark using the CSG geometry, no distinction is made between in- and ex-core regions—the defined material volumes represent the total ones in system. A single transfer rate coefficient is applied to the fuel salt to simulate the continuous removal of noble gases, specifically xenon (Xe) and krypton (Kr), from the fuel region and transfer to the off-gas system (OGS), which is not defined geometrically as not being part of the in-core active geometry. This setup allows a focused comparison of the reprocessing implementation across different depletion codes.

The removal is characterized by the following transfer rate coefficient:

$$t_{\text{fuel} \rightarrow \text{OGS}} = \frac{\dot{V}}{V_{\text{tot}}} \quad (5)$$

—where:

- V_{tot} is the total fuel volume (500 L) as reported in Table 1.
- $\dot{V}$ is the fuel salt volume flow rate, set to 3L/s.

This results in a transfer rate coefficient of 0.006 s^{-1} , assuming a removal efficiency ϵ of 1.

4.1.1 Implementation in different codes

- OpenMC: the transfer is implemented using the `transfer_rate` feature through the Python API. A continuous transfer rate coefficient of 0.006 s^{-1} is assigned to the isotopes of Xe and Kr from the fuel salt to the OGS and added to the OpenMC depletion integrator object as such:

```
integrator.add_transfer_rate(fuel,
                             ['Xe', 'Kr'],
                             0.006,
                             transfer_rate_
                             units='1/s',
                             destination_
                             material=ogs)
```

where `fuel` and `ogs` are materials defined in the OpenMC `model.xml` input file.

- Serpent: the transfer is implemented using three nested cards with hierarchic dependency. Firstly in `mflow` (mass flow), the isotopes to be transferred as well as their associated time constants are implemented. Then, in the `rep` (reprocessor) card, source and destination materials are pointed, as well as the transfer mode. Here, 1 corresponds to continuous transfer associated with artificial decay. Finally, the `dep pro` card includes the present stream into the depletion calculation. The settings following `dep pro` are conventional burnup settings.

```
mflow gasRem          % mass flow block
Xe 0.006
Kr 0.006
rep offgas             % reprocessor block
rc fuel ogs gasRem 1
dep pro offgas         % Depletion block
```

- EQL0D: in EQL0D, Matlab classes are defined for the identification of materials to be depleted and for the associated reprocessing streams. In the Matlab part of the EQL0D input file, an instance of the reprocessing stream class is created while running the input; it contains all of the required information to create material flows.

In `Rep(n)`, n is the n -th reprocessing stream, “Rep1” is the user-given reprocessor’s name, “fuel” and “ogs” are source and

² https://github.com/openmc-dev/data/blob/master/depletion/serpent_fissq.json

³ http://serpent.vtt.fi/mediawiki/index.php/Default_isomeric_branching_ratios

TABLE 2 Main parameters used to calculate transfer rate coefficients in the CAD model with external zones.

Parameter	Fuel salt	Blanket salt	D ₂ O moderator	OGS
In-core volume (cm ³)	245,700	2,025,000	2,866,100	0
Ex-core volume (cm ³)	254,300	975,000	133,900	1.00 × 10 ⁶
Volume ratio	1.035	0.481	0.0467	–
Flow rate (L/s)	3	3	50	–
Cycle time in-core (s)	81.9	675	57.3	–
Cycle time out-core (s)	84.77	325	2.68	–
λ in-core → out-core (1/s)	0.01221	0.00148	0.01745	–
λ out-core → in-core (1/s)	0.01180	0.00308	0.37341	–
λ out-core → OGS (1/s)	0.01180	0	0	–

TABLE 3 k_{eff} comparison at BOC between OpenMC and Serpent for CSG and CAD cases.

Case	OpenMC	Serpent
CSG model	1.04315 ± 0.00018	1.04333 ± 0.00023
CAD model	1.00200 ± 0.00018	1.00219 ± 0.00023

destination materials, “volatiles” is the vector of isotopes to be transferred (which can be set by ZAI identifier for single isotopes or atomic number for the transfer of all isotopes of a certain element), “eff” is an additional factor for scaling the performance of material transfers (1 for all isotopes, if left empty), 0.006 is the removal time constant, and “cont” identifies the material transfer mode—in this case, continuous transfer associated with artificial decay.

```
volatiles=[36, 54];
eff=[];
REP(n)=Rep('Rep1', 'fuel', 'ogs', volatiles,
eff, 0.006, 'cont');
```

4.2 Model 2

Aside from the geometry, the material definitions and simulation settings are identical to those used in Model 1. The purpose of this benchmark is to verify that, despite using different methods to perform Monte Carlo transport calculations on CAD geometries, the results remain consistent among the different codes.

4.3 Model 3

The final benchmark adds to the CAD geometry of Model 2 more detailed reprocessing schemes to simulate a more realistic case with transfer for all depletable inventories from the in- to ex-core regions and more accurate removal rates for noble gasses directly from the ex-core fuel region. Ex-core regions are not implemented in the geometry. A summary of the transfer coefficients adopted can be found in Table 2, where a total of seven interconnected transfer rates are defined.

As mentioned in Section 2, cyclic material transfers cannot be modeled within one single depletion step in Serpent. In this section, only OpenMC and EQL0D are therefore compared.

Moreover, as the decay transfer model is probabilistic, it can be physically interpreted as representing the statistical residence time distribution of fluid packets in a well-mixed or continuously stirred system and not a plug flow system. Therefore, it would properly represent the reactor flows only in the case in which the reactor fluids would constantly and homogeneously mix. However, as the fluid’s residence time is substantially shorter than the duration of the whole depletion step, the implementation of this model is still valid for capturing and testing the validity of the depletion calculation results in the case in which part of the reactor fluids are allowed to freely decay in an out-of-core region and to test the numerical behavior of the depletion steps once the burn-up matrices are substantially extended to mimic material transfers.

4.3.1 Implementation in different codes

- OpenMC: circulation between the in- and out-core regions is modeled by defining two corresponding transfer processes: one from in- to out-core and another from out- to in-core, for the entire inventory. For the fuel salt, this is defined as follows:

```
integrator.add_transfer_rate(fuel_in_core,
                             all_elements,
                             0.01221,
                             destination_
                             material=fuel_
                             out_core)
integrator.add_transfer_rate(fuel_out_core,
                             all_elements,
                             0.01180,
                             destination_
                             material=fuel_
                             in_core)
```

Where `fuel_in_core` and `fuel_out_core` are materials defined in the OpenMC `materials.xml` and `all_elements` is a list containing all circulating elements.

TABLE 4 Model 1: mass, activity, and radiotoxicity values from OpenMC, Serpent, and EQL0D simulations for fission products and synthetic actinides in fuel and blanket salts. Relative differences are computed with respect to OpenMC.

(a) Model 1 - Fuel salt						
Fuel - fission products - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	21.4326	1.51462E+17	1.05205E+09	99.3300	1.61900E+09	99.2285
Serpent	21.4317	1.51419E+17	1.04828E+09	99.3285	1.61316E+09	99.2267
EQL0D	21.4319	1.51429E+17	1.04830E+09	99.3281	1.61320E+09	99.2263
Fuel - fission products - relative difference [%] vs. OpenMC						
Serpent	+0.00421	+0.02786	+0.35916	+0.00148	+0.36073	0.00185
EQL0D	+0.00314	+0.02164	+0.35664	+0.00190	+0.35836	0.00226
Fuel - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	8.81666	1.70953E+16	7.08768E+06	0.66918	1.09530E+07	0.67131
Serpent	8.80440	1.70717E+16	7.07785E+06	0.67065	1.09377E+07	0.67278
EQL0D	8.81005	1.70826E+16	7.08239E+06	0.67106	1.09447E+07	0.67320
Fuel - synthetic actinides - relative difference [%] vs. OpenMC						
Serpent	+0.13875	+0.13885	+0.13652	−0.21974	+0.13957	−0.21989
EQL0D	+0.07488	+0.07457	+0.06662	−0.28110	+0.07566	−0.28139
(b) Model 1 - blanket salt						
Blanket - fission products - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	8.91484E-04	6.87242E+12	1.74676E+05	1.67641	1.71456E+06	0.99274
Serpent	8.93467E-04	6.88420E+12	1.74947E+05	1.67935	1.71776E+06	0.99464
EQL0D	8.97504E-04	6.91222E+12	1.75695E+05	1.68486	1.72553E+06	0.99888
Blanket - fission products - relative difference [%] vs. OpenMC						
Serpent	−0.21998	−0.16767	−0.15621	−0.17537	−0.17363	−0.19131
EQL0D	−0.35040	−0.57016	−0.58829	−0.50436	−0.63188	−0.61860
Blanket - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	15.4395	2.36525E+16	9.95907E+06	95.5794	3.50043E+07	20.2678
Serpent	15.4357	2.36468E+16	9.95666E+06	95.5759	3.49959E+07	20.2638
EQL0D	15.4505	2.36694E+16	9.96620E+06	95.5731	3.50294E+07	20.2780

(Continued on following page)

TABLE 4 (Continued) Model 1: mass, activity, and radiotoxicity values from OpenMC, Serpent, and EQL0D simulations for fission products and synthetic actinides in fuel and blanket salts. Relative differences are computed with respect to OpenMC.

Blanket - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
Relative difference [%] vs. OpenMC						
Serpent	0.02411	0.02468	0.02415	0.00362	0.02410	0.01986
EQL0D	−0.06991	−0.11622	−0.07072	0.00661	−0.07328	−0.05040

TABLE 5 Model 1: mass, activity, and radiotoxicity values from OpenMC, Serpent, and EQL0D simulations for fission products and synthetic actinides in OGS and moderator. Relative differences are computed with respect to OpenMC.

(a) Model 1 - OGS						
Absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT.	Inh. RT.
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	9.54979	1.71528E+16	2.67653E+09	100.0	1.77622E+10	100.0
Serpent	9.55091	1.71640E+16	2.67685E+09	100.0	1.77753E+10	100.0
EQL0D	9.52699	1.71651E+16	2.67697E+09	100.0	1.77764E+10	100.0
Relative difference [%] vs. OpenMC						
Serpent	−0.01176	−0.06531	−0.01199	0.0	−0.07392	0.0
EQL0D	+0.23665	+0.07172	−0.01641	0.0	−0.08090	0.0
(b) Model 1 - moderator						
Absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT [%]
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	6.08927E-03	2.16765E+12	91.0412	99.5937	563.588	99.9521
Serpent	6.10347E-03	2.17270E+12	91.2534	99.5746	564.902	99.9480
EQL0D	6.10425E-03	2.17298E+12	91.2651	99.5783	564.974	99.9485
Relative difference [%] vs. OpenMC						
Serpent	−0.23284	−0.23285	−0.23285	0.01916	−0.23350	0.00411
EQL0D	−0.24630	−0.24625	−0.24625	0.01547	−0.24636	0.00366

- EQL0D: also in EQL0D, the circulation between the in- and out-core regions is modeled by defining two transfer processes for the entire inventory. For the fuel salt, this is defined as follows:

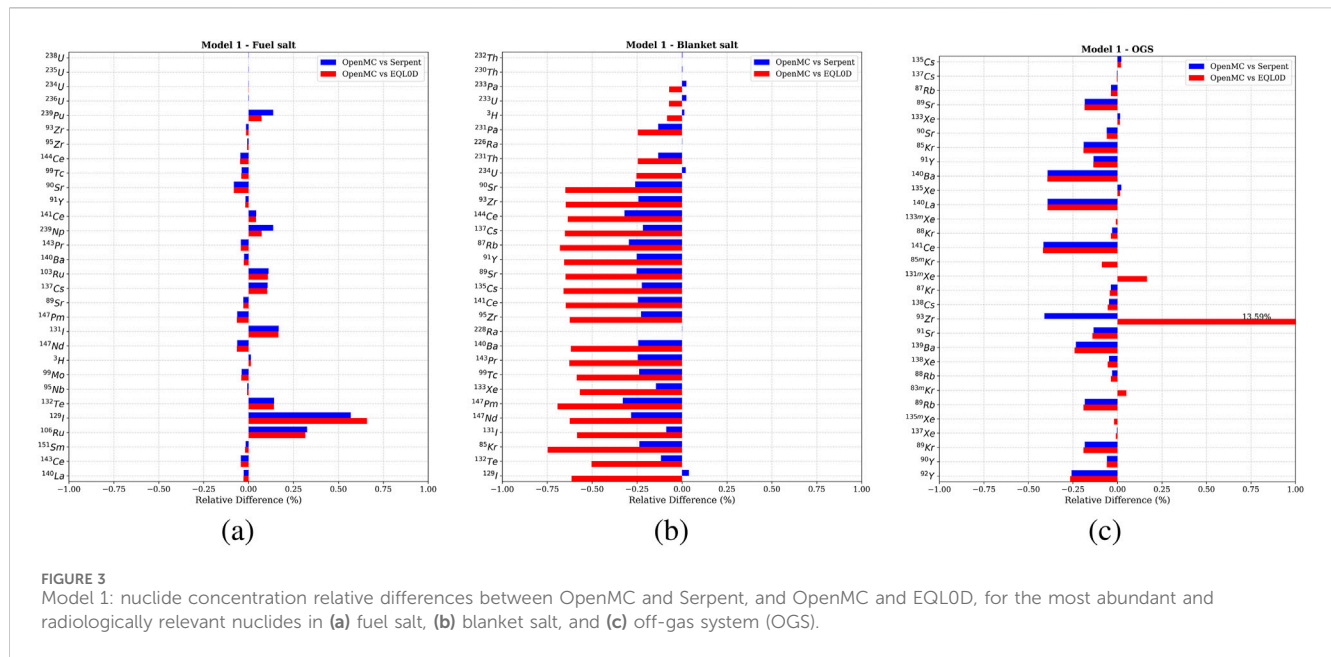
```
all_isotopes = 1:111;
REP(n)=Rep("fuel_in_out," "fuel_in," "fuel_out," all_isotopes, [], 0.01221, "cont");

REP(n+1)=Rep("fuel_out_in," "fuel_out," "fuel_in," all_isotopes, [], 0.01180, "cont");
```

Here "fuel_in" and "fuel_out" are the in- and ex-core fuel inventories, while all_isotopes is the vector including all the atomic numbers, implying that all the material isotopes are to be transferred.

5 Results

To enable a meaningful comparison of the source term and radiological impact for later analyses, the end-of-cycle (EOC) inventories are filtered to exclude radionuclides with negligible



contributions. Specifically, those lacking corresponding dose conversion coefficients for radiotoxicity estimation are excluded from the comparison. The dose coefficients are taken from the EAF (European Activation File) - 2010 Neutron-Induced Cross Section Library (Sublet et al., 2010), used as a default for ingestion and inhalation radiotoxicity for Serpent versions prior to v2.1.30. For the isotopes with dose conversion coefficients tabulated in the ICRP Publication 119 ICR (2012) and used in Serpent versions after (and including) v2.1.30, the values from the EAF were overwritten. This was done to combine an extensive tabulation of dose conversion coefficients to cover volatile fission products such as krypton and xenon (from the EAF) as well as selecting recent coefficient when possible⁴.

5.1 Neutronics comparison

The effective multiplication factors were obtained from all transport models and compared, resulting in a difference at beginning-of-cycle (BOC) of approximately 20 pcm for both cases and within statistical uncertainty (Table 3).

5.2 Model 1: CSG benchmark

Approximately 20 g of fission products and 10 g of synthetic actinides, such as plutonium in the fuel salt and ²³³U in the blanket salt, were generated in the fuel salt after 30 days of irradiation at 1 MW(th). Table 4a presents the total masses,

activities, and radiotoxicities at end-of-cycle (EOC) in the fuel salt, comparing results across the different codes. Relative differences of Serpent and EQL0D with respect to OpenMC are also reported for both fission products and synthetic actinides. The results demonstrate excellent agreement: total mass differences remain below 0.004% for fission products and 0.14% for actinides, while dose-related discrepancies are within 0.35% and 0.14%, respectively. Similarly, the total fission products and actinides in the blanket salt are reported in Table 4b. Here, fission products, although only created in small amounts in the order of ~ 10 mg, show slightly higher differences, with relative deviations below 0.2% for Serpent and 0.65% for EQL0D. In the blanket salt, transmutation via (*n*,*γ*) reactions in ²³²Th leads to the production of approximately 15 g of ²³³Pa and ²³³U combined, also showing good agreement among the codes, with relative differences below 0.72%. The remaining 10 g of fission products, primarily consisting of separated Xe and Kr along with their decay products accumulated over 30 days in the OGS, are shown in Table 5a. Relative comparisons indicate strong agreement, with all differences below 0.1% for Serpent and a slightly higher mass deviation of 0.3% for EQL0D but in line with the original FP discrepancies observed in the fuel.

Finally, OpenMC slightly underestimates tritium production in the heavy water moderator, with a difference of 0.23% (Table 5b).

Most differences in comparing nuclide concentrations in OpenMC with Serpent and EQL0D are below 0.1% and within 0.25% for the most abundant and radiological relevant nuclides in the fuel salt, with the exception of ¹²⁹I (Figure 3a). The good compatibility of fission product inventories comes from the fact that the main reaction responsible for their production is the fission of ²³⁵U, for which the fission yields are extensively historically tabulated and well-implemented in the depletion matrix by all codes. As expected, the discrepancy between synthetic actinides is

⁴ The full list of ingestion and inhalation coefficients used in the analysis is attached in the dataset.

TABLE 6 Model 2: mass, activity, and radiotoxicity values from OpenMC, Serpent, and EQL0D simulations for fission products and synthetic actinides in fuel and blanket salts. Relative differences are computed with respect to OpenMC.

(a) Model 2 - fuel salt						
Fuel - fission products - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	21.4327	1.51591E+17	1.05611E+09	99.2212	1.62468E+09	99.1196
Serpent	21.4316	1.51542E+17	1.05171E+09	99.2175	1.61787E+09	99.1155
EQL0D	21.4319	1.51551E+17	1.05172E+09	99.2176	1.61789E+09	99.1155
Fuel - fission products - relative difference [%] vs. OpenMC						
Serpent	+0.00513	+0.03231	+0.41489	+0.00371	+0.40914	+0.00415
EQL0D	+0.00373	+0.02634	+0.41382	+0.00365	+0.40893	+0.00410
Fuel - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	10.2999	1.99720E+16	8.28015E+06	0.77791	1.27955E+07	0.78064
Serpent	10.3059	1.99836E+16	8.28497E+06	0.78159	1.28030E+07	0.78434
EQL0D	10.3053	1.99825E+16	8.28449E+06	0.78154	1.28023E+07	0.78429
Fuel - synthetic actinides - relative difference [%] vs. OpenMC						
Serpent	−0.05834	−0.05802	−0.05948	−0.47315	−0.05813	−0.47501
EQL0D	−0.05325	−0.05250	−0.05412	−0.46590	−0.05377	−0.46780
(b) Model 2 - blanket salt						
Blanket - fission products - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	4.45080E−03	2.91445E+13	7.17149E+05	6.65874	7.85170E+06	4.40714
Serpent	4.44119E−03	2.90819E+13	7.15662E+05	6.65293	7.83501E+06	4.39923
EQL0D	4.42789E−03	2.90006E+13	7.13565E+05	6.63690	7.81171E+06	4.38698
Blanket - fission products - relative difference [%] vs. OpenMC						
Serpent	+0.21433	+0.21243	+0.20523	+0.08726	+0.21029	+0.17945
EQL0D	+0.51292	+0.47888	+0.49577	+0.32791	+0.49855	+0.45748
Blanket - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	15.1498	2.31997E+16	9.76696E+06	90.6864	3.43179E+07	19.2625
Serpent	15.1322	2.31725E+16	9.75550E+06	90.6890	3.42774E+07	19.2462
EQL0D	15.1266	2.31641E+16	9.75198E+06	90.7036	3.42651E+07	19.2429

(Continued on following page)

TABLE 6 (Continued) Model 2: mass, activity, and radiotoxicity values from OpenMC, Serpent, and EQL0D simulations for fission products and synthetic actinides in fuel and blanket salts. Relative differences are computed with respect to OpenMC.

Blanket - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
Blanket - synthetic actinides - relative difference [%] vs. OpenMC						
Serpent	+0.11049	+0.11721	+0.11700	−0.00292	+0.11801	+0.08473
EQL0D	+0.15287	+0.15384	+0.15300	−0.01902	+0.15374	+0.10161

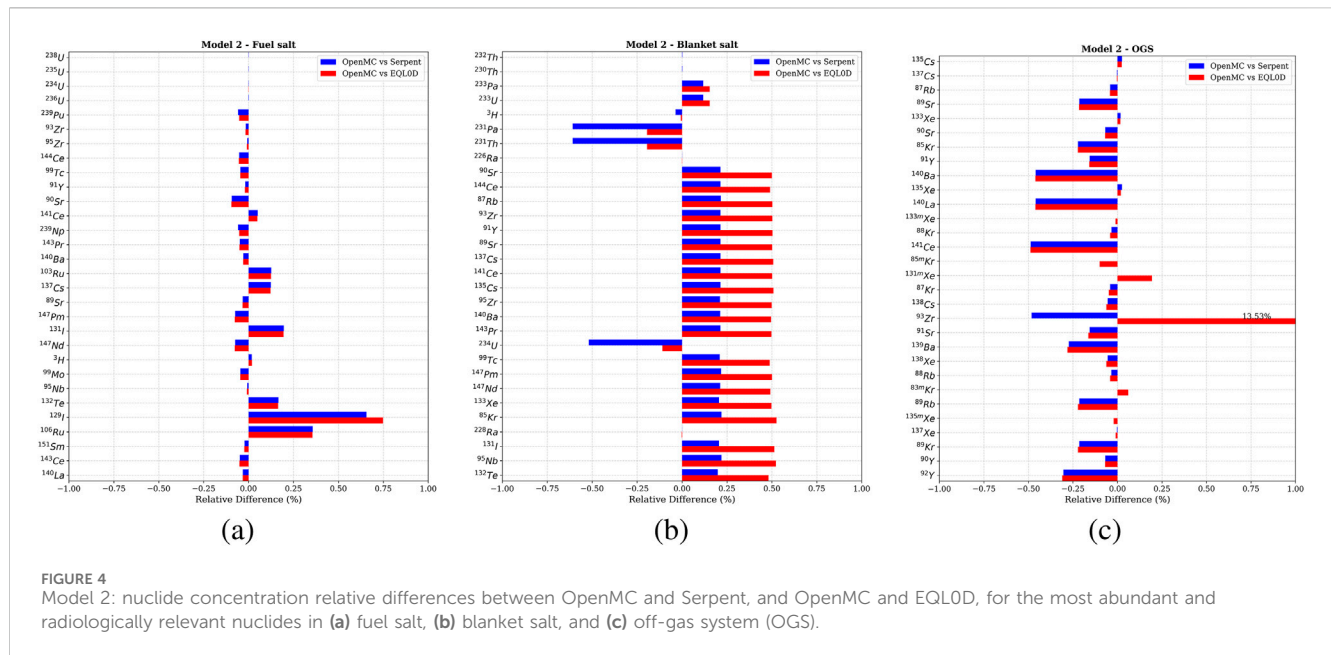
TABLE 7 Model 2: mass, activity, and radiotoxicity values from OpenMC, Serpent, and EQL0D simulations for fission products and synthetic actinides in OGS and moderator. Relative differences are computed with respect to OpenMC.

(a) Model 2 - OGS						
Absolute values						
Code	Total Mass	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	9.54498	1.71342E+16	2.67415E+9	100.0	1.77252E+10	100.0
Serpent	9.54626	1.71471E+16	2.67451E+9	100.0	1.77403E+10	100.0
EQL0D	9.52237	1.71482E+16	2.67463E+9	100.0	1.77415E+10	100.0
Relative difference [%] vs. OpenMC						
Serpent	−0.0139	−0.0754	−0.0136	0.0	−0.0854	0.0
EQL0D	+0.2365	−0.0818	−0.0181	0.0	−0.0923	0.0
(b) Model 2 - moderator						
Absolute values						
Code	Mass	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	4.49552E−03	1.60031E+12	67.2129	99.4490	416.080	99.9312
Serpent	4.50229E−03	1.60272E+12	67.3141	99.4489	416.706	99.9326
EQL0D	4.50340E−03	1.60311E+12	67.3307	99.4575	416.809	99.9336
Relative difference [%] vs. OpenMC						
Serpent	−0.1508	−0.1506	−0.1512	4.18723E−05	−0.1499	−0.00130
EQL0D	−0.1749	−0.1747	−0.1626	−0.008640762	−0.1746	−0.00236

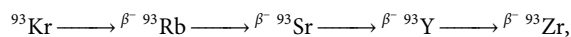
still small but slightly higher, as these generate from the transmutation chains of four different uranium isotopes present in the salt-embedded enriched uranium (^{234}U , ^{235}U , ^{236}U , ^{238}U).

In the blanket salt, fission products from bred ^{233}U appear to be systematically lower than predicted by OpenMC by approximately 0.25% and 0.65% in comparison with Serpent and EQL0D, respectively (Figure 3b). Increasing discrepancies with respect to the fuel salt are expected as a combination of reduced statistics in the blanket domain because of the possible discrepancy propagation in the breeding and ultimate fission of ^{233}U , beside the small difference in the depletion solvers to treat energy-dependent fission product yields.

Although not shown, tritium production in OpenMC is lower, estimated by about 0.25%. The main fission products removed from the fuel salt and transferred to the off-gas system (OGS) indicate good agreement between codes and validate the correct implementation of continuous removal in the Bateman equation (Figure 3c). In the OGS, the discrepancy of ^{93}Zr in the EQL0D code is notable. The initial suspect for the discrepancy was related to a different branching in the decay matrix of the two codes. Indeed, the decay matrices used in EQL0D are generated using version 2.2.0 of Serpent, which might differ from the decay matrices generated by version 2.2.1. Looking more closely, radionuclide ^{93}Zr in the OGS



inventories typically originates from the β^- decay chain of a krypton precursor. The most direct chain is



where each arrow denotes a β^- decay. Among all the intermediate nuclides, no relevant inventory discrepancies are observed before the generation of ^{93}Zr . Moreover, as ^{93}Y fully undergoes β^- decay to produce ^{93}Zr , it is confirmed that discrepancies in the isotopic concentrations cannot belong to this decay chain. The decay constant related to the decay of ^{93}Zr has likewise been checked within the EQL0D decay matrix and is consistent with the data tabulated in the presently used nuclear data library. It is thus suspected that the issue is beyond the content of the decay matrix but is related to the implementation of the numerical integration in EQL0D. The issue is currently under investigation.

5.3 Model 2: CAD benchmark

The results for the fuel salt, OGS, and heavy water closely match those from the CSG model, with the same conclusions applicable. Total masses and activity in fuel salt fall all under 0.006% for both fission products and actinides, while radiotoxicity values are within 0.4% and 0.06% for fission products and actinides, respectively (Table 6a). As shown in Table 6b, the discrepancies in fission product inventories in the blanket salt remain below 0.2% for Serpent and 0.5% for EQL0D, while actinide differences are all within 0.2%. The inventories for the OGS and heavy water are also presented in Table 7.

Despite the excellent agreement observed, as shown in Figure 4, the systematic discrepancy in the blanket salt fission products is reversed, with OpenMC exhibiting a slight

over-prediction relative to Serpent by approximately 0.2% and relative to EQL0D by about 0.5% (Figure 4b). While in the CSG model the discrepancy has to be related to the basic implementation or setup of the depletion routines, in the case of the CAD geometry, the different meshing methods for geometry implementation might have introduced a slight geometrical inconsistency, causing a relative variation in the blanket neutron flux among the codes. However, even though reversed, the blanket fission product discrepancy is still comparable to the CSG case and of limited importance since the radiotoxicities induced by blanket fission products are very limited.

5.4 Model 3: CAD with external zones benchmark

With the circulation of the inventory between in-core and out-core and the corresponding increase in the burnup block matrix size, more discrepancies arise due to numerical errors. Nevertheless, the overall agreement between OpenMC and EQL0D remains satisfactory. In the fuel salt, the total mass and activity of fission products and actinides differ by less than 0.7% and 0.2%, respectively. Radiotoxicity discrepancies are within 0.6% for fission products and 0.4% for actinides (Table 8a). The blanket salt, reported in Table 8b, shows larger deviations in low-abundance fission products, although all remain within 4%. In contrast, the more abundant actinides display good agreement, with differences below 0.7%.

In the OGS, total mass and activity differences are under 0.6% and 0.4%, respectively, while ingestion and inhalation radiotoxicity discrepancies are approximately 1% (Table 9a). Finally, all evaluated quantities for the heavy water fall within 0.2%, Table 9b.

TABLE 8 Model 3: mass, activity, and radiotoxicity values from OpenMC and EQL0D simulations for fission products and synthetic actinides in fuel and blanket salts. Relative differences are computed with respect to OpenMC.

(a) Model 3 - fuel salt						
Fuel - fission products - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	21.9240	1.53862E+17	1.06817E+09	99.2301	1.66138E+09	99.1390
EQL0D	21.7638	1.53613E+17	1.06344E+09	99.2246	1.65189E+09	99.1314
Fuel - fission products - relative difference [%] vs. OpenMC						
EQL0D	0.73096	0.16214	0.44307	0.00548	0.57115	0.00766
Fuel - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	10.2985	1.99692E+16	8.27902E+06	0.76909	1.27938E+07	0.76344
EQL0D	10.3686	2.00089E+16	8.30114E+06	0.77454	1.28392E+07	0.77049
Fuel - synthetic actinides - relative difference [%] vs. OpenMC						
EQL0D	-0.68084	-0.19879	-0.26708	-0.70778	-0.35454	-0.92328
(b) Model 3 - blanket salt						
Blanket - fission products - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	4.43291E-3	2.90336E+13	7.14423E+05	6.63281	7.82077E+06	4.39022
EQL0D	4.28075E-3	2.85510E+13	6.94844E+05	6.43555	7.65732E+06	4.29700
Blanket - fission products - relative difference [%] vs. OpenMC						
EQL0D	3.43250	1.66223	2.74050	2.97397	2.08996	2.12346
Blanket - synthetic actinides - absolute values						
Code	Mass total	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	15.1556	2.32085E+16	9.77069E+06	90.7126	3.43310E+07	19.2718
EQL0D	15.2483	2.32350E+16	9.81619E+06	90.9162	3.45554E+07	19.3912
Blanket - synthetic actinides - relative difference [%] vs. OpenMC						
EQL0D	-0.61129	-0.11409	-0.46565	-0.22448	-0.65389	-0.61946

When examining nuclide concentrations differences in Figure 5, the discrepancies are more pronounced than earlier cases. This is primarily due to the increased size of the burn-up block matrix (with seven interconnected transfers) and potential numerical instabilities. Nevertheless, the majority of fission product and actinide concentrations in the fuel salt and OGS remain within 1%. The blanket salt shows the largest

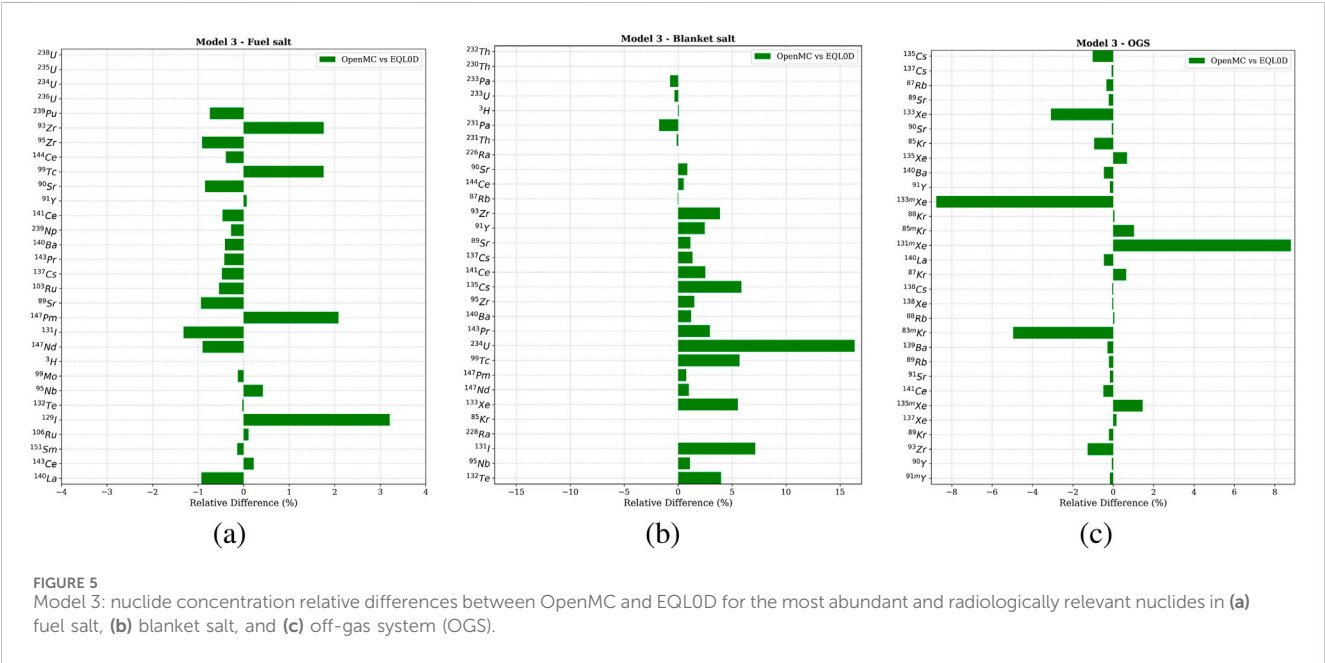
differences in fission product concentrations, though most remain below 5%.

Comparing the inventories with those from Model 2, it is noteworthy that the effect of circulation between in- and ex-core regions, along with the removal of fission products from the ex-core to the OGS, introduce a non-negligible difference in the fuel salt fission products, where more fission products are contained in

TABLE 9 Model 3: mass, activity, and radiotoxicity values from OpenMC and EQL0D simulations for fission products and synthetic actinides in OGS and moderator. Relative differences are computed with respect to OpenMC.

(a) Model 3 - OGS						
Absolute values						
Code	Total Mass	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	9.05364	1.48627E+16	2.66205E+09	100.000	1.76884E+10	100.000
EQL0D	9.00384	1.49144E+16	2.69215E+09	100.000	1.75507E+10	100.000
Relative difference [%] vs. OpenMC						
EQL0D	0.550023	−0.347855	−1.13073	0.00000	0.778238	0.00000

(b) Model 3 - moderator						
Absolute values						
Code	Mass	Activity	Ing. RT	Ing. RT	Inh. RT	Inh. RT
	[g]	[Bq]	[Sv]	[%]	[Sv]	[%]
OpenMC	4.49511E-3	1.60016E+12	67.2067	99.9999	416.042	99.9999
EQL0D	4.50262E-3	1.60283E+12	67.3190	99.4588	416.737	99.9338
Relative difference [%] vs. OpenMC						
EQL0D	−0.167045	−0.167045	−0.167045	0.541118	−0.167045	0.066086



Model 3. The same holds for the OGS system inventory, where less fission gases are present in Model 3. However, the joint contribution of fuel salt and OGS fission products inventories match the joint ones of Model 2. This is because in Model 3, the rate of fission gas extraction is based on the processing time of only the external part of the fuel salt, and the gases have been extracted from there only. Since only krypton and xenon are extracted, whenever these are generated in the in-core fuel salt through fission, a considerable amount of time is being allowed to the in-core fission products to decay before being pumped to the out-

core region and, finally, to the OGS. The fast-decaying isotopes of xenon and krypton therefore have time to be transformed into different elements through radioactive decay while being in the in-core and out-core regions before having the chance of extraction. These isotopes are not ultimately extracted to the OGS as they are no longer krypton and xenon isotopes, causing the fuel to present a higher and the OGS a lower mass of fission products, respectively.

6 Conclusion and future work

To accurately evaluate the MSE inventory and radiotoxicity at the end of the test, a comparative analysis was performed using three different depletion codes. In particular, a new Transfer Rates methodology was implemented in the OpenMC Monte Carlo transport code, similar to the approach used in Serpent and EQL0D. This feature enables the simulation of continuous fission product removal in molten salt reactors and supports the modeling of material circulation between different reactor regions. To assess the accuracy of the new depletion capability, three benchmark cases of increasing complexity were developed, incorporating progressively more advanced geometries and transfer rate features. The alignment of cross sections and nuclear data was meticulously verified by the authors, although minor inconsistencies or residual errors may still be present. The first two benchmark cases demonstrate very good agreement, with final radiotoxicity values for all inventories agreeing within 0.7% for fission products and approximately 0.2% for synthetic actinides. The third, more complex case also exhibits strong consistency; results for the fuel salt, off-gas system (OGS), and moderator remain within 1% between OpenMC and EQL0D. The only notable discrepancy is in the fission products of the blanket salt, which deviate by up to 3% but are considered insignificant due to the very small quantities involved. As expected, increasing the number of depletable materials and introducing additional transfers between them can lead to larger discrepancies, likely due to numerical instabilities in the depletion solver when handling large burnup matrices. However, we have demonstrated that for systems with up to seven interconnected materials exchanging nuclides, the depletion results remain consistent. This confirms that OpenMC can reliably estimate the nuclide inventory of the MSE.

Several simplifying assumptions were made to facilitate the comparison. In future research, a more comprehensive evaluation should include the impact of temperature interpolation in cross sections, the implementation of more advanced depletion integration algorithms, and comparisons across different nuclear data libraries.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary Material](#); further inquiries can be directed to the corresponding author.

Author contributions

LC: Conceptualization, Data curation, Formal Analysis, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review and editing. AR: Conceptualization, Data curation, Formal Analysis, Methodology, Resources, Software, Validation, Visualization, Writing – review and editing.

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Conflict of interest

Author LC was employed by Copenhagen Atomics.

The remaining author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fnuen.2025.1677436/full#supplementary-material>

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